

Student Management System

A menu-driven **C++ application** for efficiently managing student records. The project demonstrates fundamental concepts of C++ programming, including object-oriented programming, file handling, data validation, and CRUD (Create, Read, Update, Delete) operations.

Features

- Add new student records
- View all student records
- Search for a student by ID
- Update existing student information
- Delete student records
- File handling for persistent data storage
- Input validation for reliable data entry
- Simple, user-friendly menu-driven interface

Technologies Used

- C++
- Standard Template Library (STL)
- File Handling (`fstream`)
- Object-Oriented Programming (OOP)

Getting Started

Prerequisites

- A C++ compiler such as **g++**, **MSVC**, or **MinGW**
- Any C++ IDE (Visual Studio Code, Code::Blocks, Dev-C++, etc.)

Compilation

```
g++ main.cpp -o StudentManagementSystem
```

Run

```
./StudentManagementSystem
```

(On Windows)

StudentManagementSystem.exe

Project Structure

```
Student-Management-Systems/  
├── main.cpp  
├── README.md  
├── LICENSE  
└── .gitignore
```

Learning Objectives

This project helps beginners understand:

- Classes and Objects
- File Input/Output
- Data Validation
- Functions
- Menu-Driven Programming
- CRUD Operations
- Basic Software Design

Future Enhancements

- Password-protected login system
- Student attendance management
- Grade and GPA calculation
- Database integration (MySQL/SQLite)
- Graphical User Interface (GUI)
- Export records to CSV or PDF

License

This project is licensed under the MIT License.

Author

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